



PSAC 2026

Mathematics

Specimen Paper 4 of 4 | Prepared by Noukclass.com | Topics: Number Patterns • Word Problems • 3D Shapes • Symmetry & Coordinates

Time allowed: 1 hour 45 minutes Total Marks: 99

Centre Number: _____

Index Number: _____

Candidate's Name: _____

INSTRUCTIONS TO CANDIDATES

1. This paper contains **45** questions in three sections.
2. Write your Index Number and Name in the spaces provided above.
3. Answer **all** questions.
4. Write your answers in the spaces provided in this booklet.
5. Show **all** working clearly. Marks may be awarded for working even if the final answer is wrong.
6. Do **not** use red, green or black ink.
7. Diagrams are **not** drawn to scale unless stated otherwise.
8. Calculators are **not** allowed.

Section	Questions	Marks Available	Marks Awarded	Examiner
A	1–18	36		
B	19–28	20		
C	29–45	43		
Total		99		

SECTION A — Short Answer Questions

Questions 1–18 | 36 marks | Write your answers clearly in the spaces provided.

1. Write the next **two terms** in this sequence: 3 7 11 2 marks

15 $\triangle$ $\triangle$

Next two terms: _____ and _____

Rule: _____

2. Write the next **two terms**: 2 6 18 54 $\triangle$ 2 marks

$\triangle$

Next two terms: _____ and _____

3. What is the **10th term** of the sequence: 5, 10, 15, 20, 25, ...? 2 marks
What is the **rule** for the n th term?

10th term: _____ n th term: _____

4. Find the missing numbers in this pattern: 1 4 9 16 $\triangle$ 36 $\triangle$ 2 marks

Missing numbers: _____ and _____

5. Name this 3D shape: it has **6 faces**, all of which are rectangles. 2 marks
How many **edges** and **vertices** does it have?

Shape: _____ Edges: _____ Vertices: _____

6. How many faces, edges and vertices does a **triangular prism** have? 2 marks

Faces: _____ Edges: _____ Vertices: _____

7. A shape has **rotational symmetry of order 4**. Draw or describe what this means. Give an example of a shape with this property. 2 marks

Example shape: _____

8. A point P is at coordinate **(3, 5)**. It is reflected in the y-axis. What are the new coordinates? 2 marks
Then it is reflected in the x-axis. What are those coordinates?

Reflected in y-axis: _____ Reflected in x-axis:

9. A box of 48 chocolates is shared equally among 6 children. 2 marks
Each child then gives **3 chocolates** back. How many does each child have now?

Answer: _____ chocolates

10. A train travels at **80 km/h**. How far does it travel in **2 hours 30 minutes**? 2 marks

Distance = _____ km

11. Find the missing term in this sequence and explain the pattern: 2 marks

100 97 91 82 $\triangle$ 55

Missing term: _____

12. How many **faces** does a **sphere** have? How many **vertices** does a **cone** have? 2 marks

Sphere faces: _____ Cone vertices: _____

13. Akil buys **4 notebooks** at Rs 35 each and **3 pens** at Rs 12.50 each. He pays with a Rs 200 note. How much change does he receive? 2 marks

Change = Rs _____

14. A shop sells T-shirts in **3 colours** (red, blue, green) and **2 sizes** (S, L). List **all possible combinations** using a systematic list. 2 marks

Total combinations: _____

15. Write the first **five terms** of a sequence where the first term is **3** and each term is **double the previous term**. 2 marks

Sequence: _____

16. A square has **4 lines of symmetry** and rotational symmetry of order **4**. How many lines of symmetry does a regular octagon have? What is its order of rotational symmetry? 2 marks

Lines of symmetry: _____ Order of rotation: _____

17. A square pyramid has a square base and **4 triangular faces**.

2 marks

Complete: Faces = _____, Edges = _____, Vertices = _____.

18. A car park has **8 rows** with **15 spaces** per row. After **74 cars** park, how many spaces are left?

2 marks



Spaces left: _____

SECTION B — Multiple Choice Questions

Questions 19–28 | 20 marks | Tick (✓) ONE box for each question. Each question is worth 2 marks.

Questions 19–20

19. The 8th term of the sequence 4, 8, 12, 16, ... is:

- ☐ A 28
- ☐ B 32
- ☐ C 36
- ☐ D 40

20. A cube has how many edges?

- ☐ A 8
- ☐ B 10
- ☐ C 12
- ☐ D 6

Questions 21–22

21. An isosceles triangle reflected in its line of symmetry produces:

- ☐ A A different triangle
- ☐ B A rectangle
- ☐ C The same triangle
- ☐ D A square

22. Pens cost Rs 15 each. How many can be bought for Rs 200?

- ☐ A 12
- ☐ B 13
- ☐ C 14
- ☐ D 15

Questions 23–24

23. Point (4, -3) is reflected in the x-axis. The new point is:

- ☐ A (-4, 3)
- ☐ B (4, 3)
- ☐ C (-4, -3)
- ☐ D (3, -4)

24. The rule for sequence 2, 5, 8, 11, ... is:

- ☐ A Multiply by 2
- ☐ B Add 3
- ☐ C Add 4
- ☐ D Subtract 2

Questions 25–26

25. A cylinder has how many curved faces?

- ☐ A 0
- ☐ B 1
- ☐ C 2
- ☐

26. Leon earns Rs 18,000 per month. He saves Rs 4,500. What fraction of his earnings does he save?

- ☐ A $\frac{1}{4}$
- ☐ B $\frac{1}{3}$
- ☐

☐ D 3

☐ C $\frac{1}{5}$

☐ D $\frac{2}{7}$

Questions 27–28

27. A sequence starts at 100 and halves each time. The 4th term is:

☐ A 25

☐ B 50

☐ C 12.5

☐ D 20

28. What is the total number of faces on two tetrahedra (triangular pyramids, each with 4 faces)?

☐ A 6

☐ B 8

☐ C 10

☐ D 12

SECTION C — Structured Questions

Questions 29–45 | 43 marks | Show ALL working. Marks are awarded for method as well as the answer.

29. The table shows the number of visitors to a museum over 5 days.

3 marks

Day	Mon	Tue	Wed	Thu	Fri
Visitors	145	210	178	230	197

- (a) What is the total number of visitors for the 5 days?

Total = _____

- (b) Find the mean number of daily visitors.

Mean = _____

- (c) On which day were there most visitors? How many more than Monday?

Day: _____ More than Monday: _____

30. Look at this pattern of squares made from tiles:

3 marks

Pattern 1 uses 1 tile. Pattern 2 uses 4 tiles. Pattern 3 uses 9 tiles.

- (a) How many tiles does Pattern 5 use?

Answer: _____ tiles

- (b) Write a formula for the number of tiles in Pattern n .

- (c) Which pattern uses 100 tiles?



Pattern number: _____

31. A baker makes **12 loaves** every hour. He starts at **05:30** and works for **6 hours**. How many loaves does he make? If he sells them for Rs 45 each, how much does he earn?

2 marks

Loaves: _____ Earnings: Rs _____

32. A point is at **(2, 3)** on a coordinate grid.

3 marks

(a) Translate the point 4 units right and 2 units down. Write the new coordinates.

New point: _____

(b) The original point (2, 3) is reflected in the line $x = 0$ (the y-axis). Write the new coordinates.

Reflected point: _____

(c) A rectangle has vertices at (1, 1), (5, 1), (5, 4) and (1, 4). Find its area.

Area = _____ square units

33. A family buys **2 adult tickets** at Rs 350 each and **3 child tickets** at Rs 175 each for a theme park. They have a **10% discount voucher**. What do they pay after the discount?

2 marks

Total after discount = Rs _____

34. The n th term of a sequence is given by the formula $3n + 2$.

3 marks

(a) Find the 1st, 2nd and 5th terms.

1st = _____ 2nd = _____ 5th = _____

(b) Which term has the value 47?

Term number: _____

(c) Is 100 a term in this sequence? Explain how you know.

35. Priya and Leon together have **Rs 840**. Priya has **Rs 120 more** than Leon. How much does each person have?

2 marks

Priya: Rs _____ Leon: Rs _____

36. Describe the 3D shape that can be made from this net: a rectangle with two semicircles attached to its longer sides.

2 marks

Shape: _____

How many edges does a **triangular pyramid (tetrahedron)** have?

Edges: _____

37. Mia saves money every week. In week 1 she saves Rs 10, in week 2 she saves Rs 20, in week 3 she saves Rs 40 (each week she doubles the previous amount).

3 marks

(a) How much does she save in week 5?

Week 5: Rs _____

(b) How much has she saved in total after 5 weeks?

Total: Rs _____

(c) In which week will she first save more than Rs 500?

Week: _____

38. A school timetable has **6 lessons** per day, each lasting **45 minutes**. Breaks total **30 minutes** per day. School runs **5 days** per week.

3 marks

(a) How many minutes of lessons per day?

Answer: _____ minutes

(b) How many hours of lessons per week?

Answer: _____ hours _____ minutes

(c) School starts at 08:00. What time does it end if total school day = lessons + breaks?

End time: _____

39. A shape has **rotational symmetry of order 3**. Explain what this means and give an example. 2 marks

Example: _____

40. A shop is having a sale. All prices are reduced by **25%**. The original prices are: 3 marks

Item	Original Price	Sale Price
Bag	Rs 480	
Shoes	Rs 1,200	
Hat	Rs 160	

- (a) Complete the sale price column.
- (b) What is the total saving if all three items are bought?

Total saving = Rs _____

41. Look at this staircase pattern made from cubes: 3 marks

Step 1: 1 cube • Step 2: 3 cubes • Step 3: 6 cubes • Step 4: 10 cubes

- (a) How many cubes in Step 5?

Answer: _____

- (b) What is the rule for finding each term (the differences between consecutive terms)?

- (c) How many cubes in total for steps 1 to 5?

Total = _____

42. Three children have a total of **Rs 2,100**. Asha has **twice** as much as Ben. Carmen has **Rs 300**. How much does each of Asha and Ben have?

2 marks

Ben: Rs _____ Asha: Rs _____

43. A container in the shape of a cuboid has a square base of side **10 cm** and a height of **15 cm**.

3 marks

- (a) Find the volume of the container.

Volume = _____ cm^3

- (b) Water fills the container to a height of **12 cm**. What volume of water is in the container?

Volume of water = _____ cm^3

- (c) What fraction of the container is empty? Give your answer in simplest form.

Empty fraction = _____

44. Describe fully the symmetry of a regular hexagon (include lines of symmetry and order of rotational symmetry).

2 marks

45. A market stall sells apples at **Rs 8 each** or **Rs 90 for a bag of 12**. Kiran needs **36 apples**. What is the cheapest way to buy them, and how much does it cost?

2 marks

Cheapest method: _____

Cost: Rs _____

Mark Scheme — Noukclass.com Specimen **Paper 4 of 4 — 2026**

This mark scheme is provided for self-checking and teacher reference only. For method marks, show all working. Prepared by Noukclass.com based on MES syllabus and past paper analysis. Not an official MES document.

Q	Answer	Marks	Notes
1	19, 23; Rule: add 4 each time	2	Arithmetic sequence with common difference 4
2	162, 486; Rule: multiply by 3	2	Geometric sequence; $54 \times 3 = 162$
3	10th term = 50; nth term = $5n$	2	Multiples of 5; $5 \times 10 = 50$
4	25 and 49	2	Square numbers: $5^2 = 25$, $7^2 = 49$
5	Cuboid; Edges=12; Vertices=8	2	Cuboid: 6 rectangular faces, 12 edges, 8 vertices
6	Faces=5; Edges=9; Vertices=6	2	Triangular prism: 2 triangles + 3 rectangles
7	It fits exactly onto itself 4 times in a 360° turn; Example: Square	2	Accept: regular 4-sided polygon, square
8	Reflected in y-axis: $(-3, 5)$; Reflected in x-axis: $(3, -5)$	2	y-axis reflection: negate x; x-axis reflection: negate y
9	5 chocolates	2	$48 \div 6 = 8$; $8 - 3 = 5$
10	200 km	2	$2.5 \text{ hours} \times 80 = 200 \text{ km}$
11	70; Pattern: subtract 3, then 6, then 9, then 12, then 15 (triangular numbers)	2	Differences: -3, -6, -9, -12, -15; $82 - 12 = 70$
12	Sphere faces=1 (curved); Cone vertices=1	2	Sphere has one curved surface; cone has 1 apex vertex
13	Rs 37.50	2	$4 \times 35 = 140$; $3 \times 12.5 = 37.5$; $140 + 37.5 = 177.5$; $200 - 177.5 = 22.5$... wait: $4 \times 35 = 140 + 3 \times 12.5 = 37.5 = 177.50$; $200 - 177.50 = 22.50$

14	Red-S, Red-L, Blue-S, Blue-L, Green-S, Green-L; Total: 6	2	3 colours $\times$ 2 sizes = 6 combinations
15	3, 6, 12, 24, 48	2	Each term = previous $\times$ 2
16	8 lines of symmetry; Order 8	2	Regular octagon: as many lines as sides
17	Faces=5; Edges=8; Vertices=5	2	Square pyramid: 1 square base + 4 triangles
18	46 spaces left	2	$8 \times 15 = 120$ total; $120 - 74 = 46$
19	B — 32	2	$4 \times 8 = 32$
20	C — 12	2	Cube: 12 edges
21	C — The same triangle	2	Reflection in line of symmetry maps shape to itself
22	B — 13	2	$200 \div 15 = 13.33$; can buy 13 (with Rs 5 change)
23	B — (4, 3)	2	Reflection in x-axis: y changes sign; (4, -3) $\rightarrow$ (4, 3)
24	B — Add 3	2	2, 5, 8, 11: difference is +3 each time
25	B — 1	2	Cylinder has 1 curved face and 2 flat circular faces
26	A — $\frac{1}{4}$	2	$4500/18000 = \frac{1}{4}$
27	C — 12.5	2	100, 50, 25, 12.5 (halving each time)
28	B — 8	2	2 tetrahedra $\times$ 4 faces = 8
29	(a) 960 (b) 192 (c) Thursday, 85 more	3	$145 + 210 + 178 + 230 + 197 = 960$; $960 \div 5 = 192$; $230 - 145 = 85$
30	(a) 25 tiles (b) n^2 (c) Pattern 10	3	$5^2 = 25$; $n^2 = 100 \rightarrow n = 10$
31	72 loaves; Rs 3,240	2	$12 \times 6 = 72$; $72 \times 45 = 3240$
32	(a) (6, 1) (b) (-2, 3) (c) 12 square units	3	$(2+4, 3-2) = (6, 1)$; reflect: (-2,3); length=4, width=3, A=12
33	Rs 945	2	$2 \times 350 + 3 \times 175 = 700 + 525 = 1050$; 10% off=105; $1050 - 105 = 945$
34	(a) 5, 8, 17 (b) 15th term (c) No; $3n+2=100 \rightarrow$ $n=32.67$, not a whole number	3	$3(1)+2=5$; $3(2)+2=8$; $3(5)+2=17$; $3n+2=47 \rightarrow 3n=45 \rightarrow n=15$
35	Priya Rs 480; Leon Rs 360	2	$L+(L+120)=840$; $2L=720$; $L=360$; $P=480$

36	Cylinder; 6 edges	2	Tetrahedron: 4 faces, 6 edges, 4 vertices
37	(a) Rs 160 (b) Rs 310 (c) Week 6 (Rs 320)	3	Wk1=10,2=20,3=40,4=80,5=160,6=320; total 5wks=10+20+40+80+160=310
38	(a) 270 min (b) 22 hours 30 min/week (c) 12:30	3	$6 \times 45 = 270$ min; $270 \times 5 = 1350$ min = 22h30m; $270 + 30 = 300$ min per day; 08:00+5h=13:00... 300min=5h; 08:00+5h=13:00. Wait: 08:00+5hours=13:00. But total school day (lessons+breaks)= $270 + 30 = 300$ min=5h; end=13:00
39	It maps to itself 3 times in a full rotation; Example: equilateral triangle	2	Order 3 means fits on itself at 120° , 240° and 360°
40	(a) Rs 360, Rs 900, Rs 120 (b) Total saving = Rs 460	3	25% off: $480 \times 0.75 = 360$; $1200 \times 0.75 = 900$; $160 \times 0.75 = 120$; saving = $120 + 300 + 40 = 460$
41	(a) 15 (b) Add 2 more each time (differences: +2,+3,+4,+5,+6) (c) 35 total	3	Triangular numbers: 1,3,6,10,15; $1+3+6+10+15=35$
42	Ben: Rs 600; Asha: Rs 1,200	2	Asha+Ben+300=2100; $2B+B=1800$; B=600; A=1200
43	(a) 1,500 cm ³ (b) 1,200 cm ³ (c) 1/5	3	$10 \times 10 \times 15 = 1500$; $10 \times 10 \times 12 = 1200$; empty = $300/1500 = 1/5$
44	6 lines of symmetry; rotational symmetry of order 6	2	Regular hexagon: n lines and order n for n-sided regular polygon
45	Buy 3 bags of 12 (=36 apples) for Rs 270; cheaper than $36 \times \text{Rs } 8 = \text{Rs } 288$	2	$3 \times 90 = 270$; $36 \times 8 = 288$; bags are cheaper